

Practical Demo: Constructing a Benchmark-Aware Bond Portfolio

From an investable universe to weights, trades, and stress tests

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This reading copy shows the calculations' results and figures. The HTML demo contains the full code in folded blocks and interactive scenario controls.

Goal and Decision

An investment-grade manager has **\$50 million** to invest. The mandate is to earn more carry than a broad bond benchmark without taking an unintended duration, curve, sector, issuer, or liquidity bet. This demo turns that mandate into an auditable portfolio and an implementable trade list.

The security descriptions and market analytics are an **illustrative teaching universe**, not dated observations or executable quotes. In practice, the same table would come from a portfolio system, evaluated pricing service, dealer runs, and reference-data checks.

Workflow stage	In this case
Supplied	Bond terms, yields, spreads, durations, liquidity costs, benchmark and current weights
Constructed	A transparent cell-sampled portfolio and a constrained optimized portfolio
Diagnosed	Factor mismatches, tracking error, concentration, turnover, and transaction cost
Used for	Trade sizing, scenario P&L, and a portfolio recommendation

1 Define the Mandate

The optimized portfolio must be fully invested, hold no more than 18% in one security, keep at least 25% in Treasuries, cap corporate exposure at 45%, and stay within 0.05 years of benchmark duration. These are teaching constraints; an actual mandate would also specify ratings, currencies, derivatives, cash, tax, ESG, and counterparty rules.

2 Assemble the Investable Universe

Yield and spread are annual percentages expressed as decimals. Duration and spread duration are in years; transaction cost is a one-way estimate in basis points of market value. KRD 2, KRD 5, and KRD 10 are simplified key-rate duration contributions per dollar invested.

	Sector	Cell	Yield
Bond			
UST 2Y	Treasury	Gov 1-4	0.0415
UST 5Y	Treasury	Gov 5-8	0.0435

	Sector	Cell	Yield
Bond			
UST 10Y	Treasury	Gov 9+	0.0455
Agency 3Y	Agency	Gov 1-4	0.0442
Agency 7Y	Agency	Gov 5-8	0.047
Bank A 3Y	Corporate	Corp 1-4	0.0505
Bank B 5Y	Corporate	Corp 5-8	0.0528
Utility A 7Y	Corporate	Corp 5-8	0.0555
Industrial A 10Y	Corporate	Corp 9+	0.0585
Industrial B 5Y	Corporate	Corp 5-8	0.054
Telecom A 10Y	Corporate	Corp 9+	0.06
Consumer A 7Y	Corporate	Corp 5-8	0.0562

	Spread (bp)	Duration	Spread duration
Bond			
UST 2Y	0	1.88	0
UST 5Y	0	4.45	0
UST 10Y	0	7.85	0
Agency 3Y	38	2.72	2.6
Agency 7Y	55	5.82	5.65
Bank A 3Y	92	2.65	2.55
Bank B 5Y	112	4.35	4.2
Utility A 7Y	138	5.65	5.45
Industrial A 10Y	165	7.35	7.05
Industrial B 5Y	125	4.18	4.02
Telecom A 10Y	182	7.1	6.75
Consumer A 7Y	148	5.45	5.25

	Trading cost (bp)	Benchmark weight
Bond		
UST 2Y	1	0.1
UST 5Y	2	0.13
UST 10Y	3	0.1
Agency 3Y	3	0.09
Agency 7Y	5	0.1
Bank A 3Y	6	0.07
Bank B 5Y	7	0.09
Utility A 7Y	9	0.08
Industrial A 10Y	12	0.07

	Trading cost (bp)	Benchmark weight
Bond		
Industrial B 5Y	8	0.07
Telecom A 10Y	14	0.05
Consumer A 7Y	10	0.05

3 Translate Holdings into Portfolio Exposures

i Read one row before constructing weights

Putting 10% of \$50 million in the UST 5Y means \$5 million of market value. Its 4.45-year duration contributes $\$5,000,000 \times 4.45 \times 0.0001 = \text{\$2,225 per bp}$. Its contribution to portfolio duration is $0.10 \times 4.45 = 0.445$ years. Dollar exposure adds across positions; security duration must first be weighted. The weighted average of individual yields is a carry proxy, not the portfolio's cash-flow internal rate of return.

For weights w_i and security exposure x_{ik} , portfolio exposure to factor k is $x_{P,k} = \sum_i w_i x_{ik}$. Dollar value of one basis point is $DV01 = V_P D_P(0.0001)$.

Yield (%)	4.944
Duration	4.842
Convexity	31.455
Spread duration	3.166
Treasury weight (%)	33.000
Corporate weight (%)	48.000
Largest position (%)	13.000
KRD 2	0.722
KRD 5	2.272
KRD 10	1.847
DV01 (\$/bp)	24209.500
dtype: float64	

4 Build a Transparent Cell-Sampled Portfolio

Cell sampling first matches broad sector/maturity buckets, then chooses a representative liquid bond in each cell. It is easy to explain, but matching cell weights does not guarantee matching duration or key-rate exposure.

	Benchmark	Cell sampled
Bond		
UST 2Y	0.1	0.19
UST 5Y	0.13	0.23
UST 10Y	0.1	0.1
Agency 3Y	0.09	0
Agency 7Y	0.1	0
Bank A 3Y	0.07	0.07
Bank B 5Y	0.09	0.29
Utility A 7Y	0.08	0
Industrial A 10Y	0.07	0.12
Industrial B 5Y	0.07	0
Telecom A 10Y	0.05	0
Consumer A 7Y	0.05	0

5 Construct a Constrained Active Portfolio

Let $a = w - w_B$ be active weights and B the factor-exposure matrix. A compact factor-risk approximation is

$$TE^2 \approx (B'a)' \Sigma_F (B'a) + \sum_i a_i^2 \sigma_{\epsilon,i}^2.$$

The objective balances predicted tracking error, turnover, transaction cost, and a modest preference for carry:

$$\min_w TE^2 + \lambda_T \sum_i (w_i - w_i^{current})^2 + \lambda_C \sum_i c_i |w_i - w_i^{current}| - \lambda_Y w' y.$$

This is a teaching risk model. Production construction would use a validated factor covariance matrix, specific-risk forecasts, executable costs, lot sizes, and compliance-tested constraints.

6 Compare the Alternatives

Check constraints before comparing yield. The cell sample is deliberately transparent, not constrained: putting a whole cell into one representative can breach the 18% security limit. The optimizer searches among feasible portfolios; it does not turn a higher assumed yield into a free excess return.

	Benchmark	Current	Cell sampled
Yield (%)	4.944	4.962	4.831
Duration	4.842	4.61	4.495
Convexity	31.455	28.776	27.932
Spread duration	3.166	3.187	2.242
Treasury weight (%)	33	30	52
Corporate weight (%)	48	52	48
Largest position (%)	13	12	29
KRD 2	0.722	0.802	0.705
KRD 5	2.272	2.235	2.17
KRD 10	1.847	1.572	1.62
DV01 (\$/bp)	24,209.5	23,048	22,473.5
Ex ante tracking error (%)	0	0.247	0.694
Turnover (%)	11	0	48
Estimated trading cost (\$)	7,250	0	30,750

	Optimized
Yield (%)	4.915
Duration	4.792
Convexity	31.024
Spread duration	3.068
Treasury weight (%)	34.805
Corporate weight (%)	45
Largest position (%)	12.416
KRD 2	0.732
KRD 5	2.278
KRD 10	1.781
DV01 (\$/bp)	23,959.5
Ex ante tracking error (%)	0.104
Turnover (%)	7
Estimated trading cost (\$)	3,462.414

7 Turn Weights into Trades

Trades below are dollar market values. A real order list would convert market value to face amount using clean price, accrued interest, minimum denominations, round lots, and available dealer inventory.

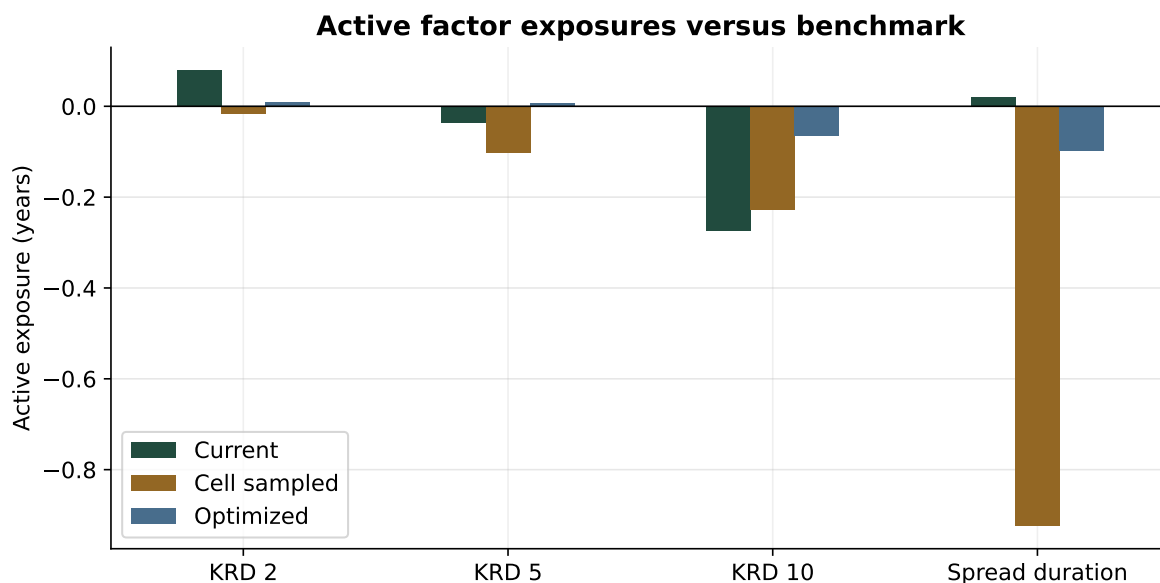


Figure 1: Optimization controls factor mismatches that cell weights alone can conceal

	Sector	Cell	Current weight (%)
Bond			
UST 2Y	Treasury	Gov 1-4	12
UST 5Y	Treasury	Gov 5-8	10
UST 10Y	Treasury	Gov 9+	8
Agency 3Y	Agency	Gov 1-4	10
Agency 7Y	Agency	Gov 5-8	8
Bank A 3Y	Corporate	Corp 1-4	8
Bank B 5Y	Corporate	Corp 5-8	10
Industrial B 5Y	Corporate	Corp 5-8	8

	Target weight (%)	Trade (\$mm)	Estimated cost (\$)
Bond			
UST 2Y	12.416	0.208	20.778
UST 5Y	11.36	0.68	136.018
UST 10Y	11.029	1.515	454.369
Agency 3Y	10.38	0.19	57.047
Agency 7Y	9.815	0.907	453.707
Bank A 3Y	4.341	-1.83	1,097.806

Bond	Target weight (%)	Trade (\$mm)	Estimated cost (\$)
Bank B 5Y	8.128	-0.936	655.032
Industrial B 5Y	6.531	-0.735	587.629

8 Stress-Test the Decision

For Treasury yield shift Δy_i and spread change Δs_i ,

$$\frac{\Delta P_i}{P_i} \approx -D_i \Delta y_i - S D_i \Delta s_i + \frac{1}{2} C_i (\Delta y_i)^2.$$

Scenario P&L (\$mm)	Benchmark	Current	Cell sampled
Parallel rates +50 bp	-1.191	-1.134	-1.106
Bear steepener	-1.515	-1.411	-1.375
Bull flattener	1.657	1.535	1.508
Credit selloff	-1.284	-1.326	-1.121

Scenario P&L (\$mm)	Optimized
Parallel rates +50 bp	-1.179
Bear steepener	-1.482
Bull flattener	1.624
Credit selloff	-1.225

8.1 Explore rate and spread shocks (HTML)

The nonparallel scenarios use each bond's three key-rate exposures, rather than assigning one maturity bucket to the entire bond. Parallel scenarios include convexity; twists use first-order key-rate risk because a full cross-key-rate convexity matrix is not supplied. Agency spreads move by one quarter of the corporate spread shock in both the scenario table and the explorer.

The table above is the static PDF fallback. In HTML, change the parallel Treasury-rate and corporate-spread shocks to stress the optimized portfolio.

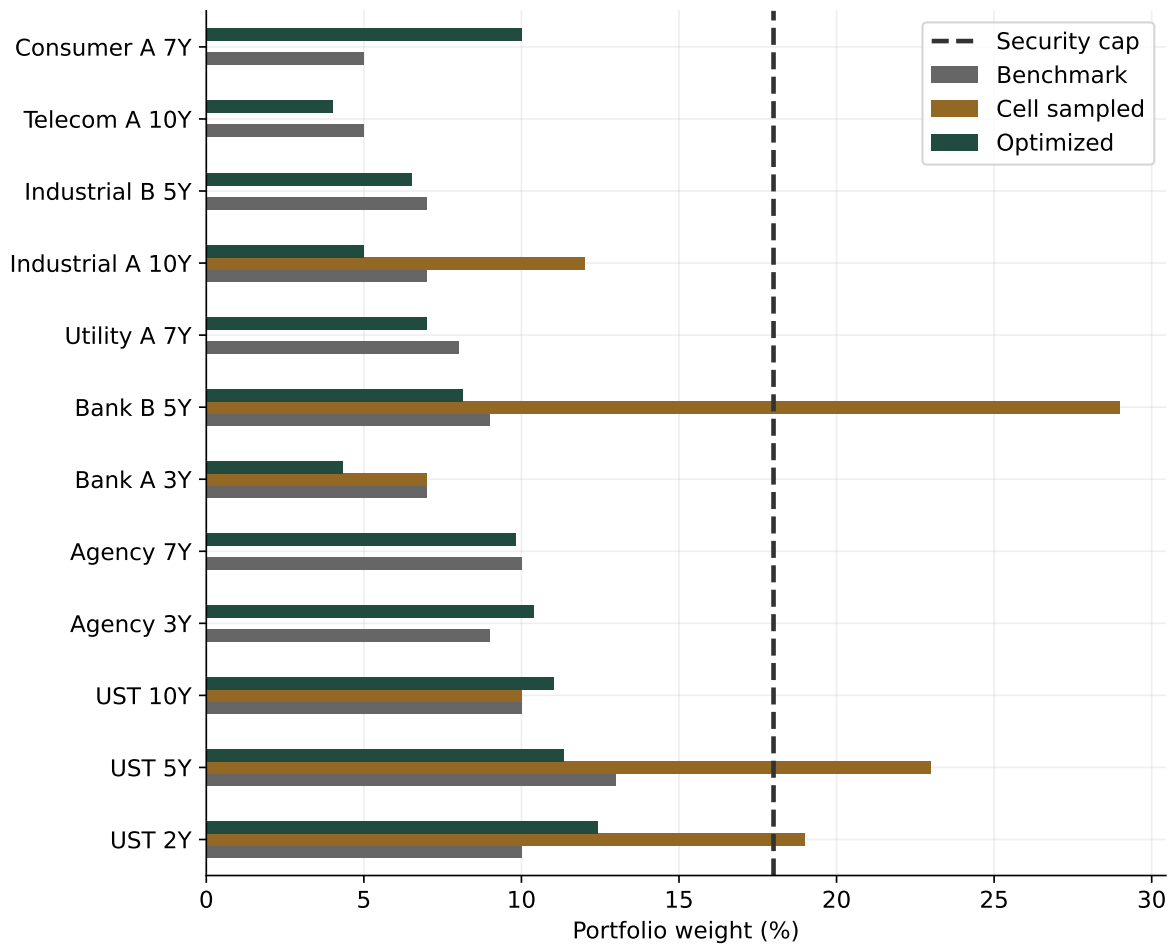


Figure 2: Security weights reveal concentration that an aggregate duration can hide.

9 Validation and Recommendation

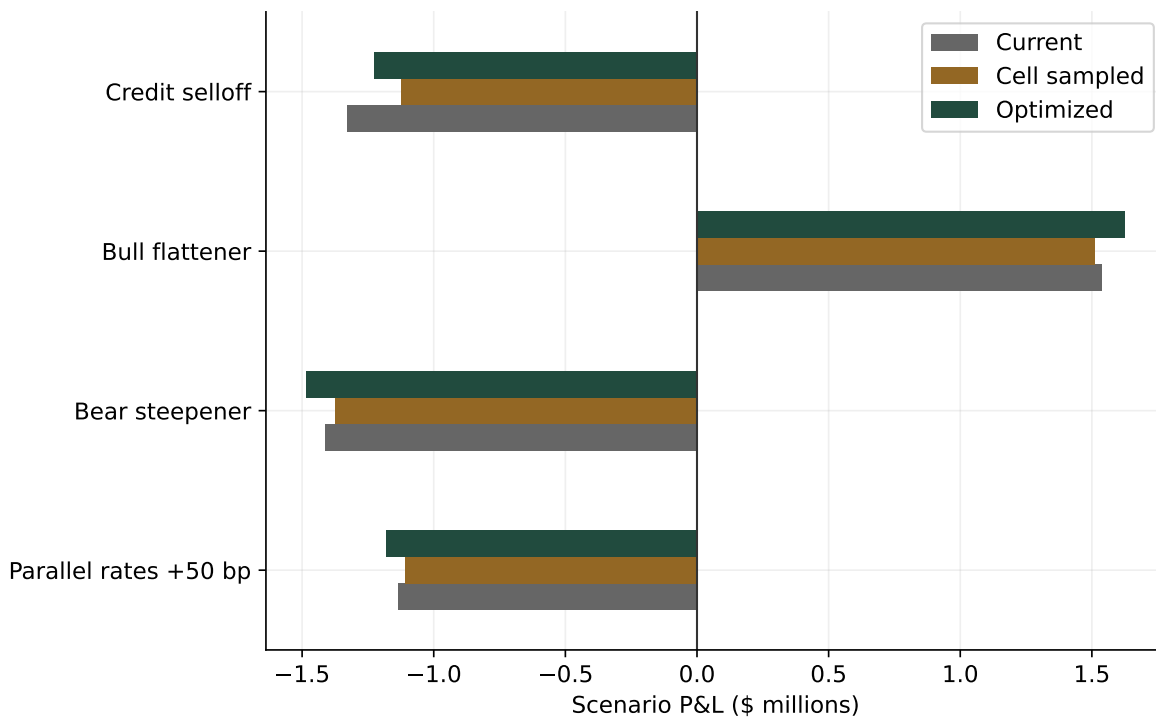


Figure 3: Compare stress losses before approving the trade list; these are scenarios, not return forecasts.

Optimization converged: True

Weights sum to	1.0000
Duration difference vs benchmark	-0.0500
Treasury minimum slack	0.0980
Corporate cap slack	0.0000
Position-cap slack	0.0558
dtype: float64	

The optimizer is a disciplined starting point, not an automatic trade instruction. The manager should prefer it to the cell sample only if its incremental carry and factor control survive executable-price checks, lot rounding, issuer review, and scenario analysis. Rebalance when the benefit of restoring the mandate exceeds transaction costs and operational complexity.

10 Instructor Extension

💡 Make a recommendation, not just a weight vector

Lower the corporate cap from 45% to 35%, rerun from the top, and compare yield, trading cost, tracking error, and the credit-selloff outcome. Explain which constraint forced the trades. Would the improvement justify the transaction cost? Keep the covariance matrix fixed so the comparison isolates the mandate change.

For a market-based extension, inspect bond terms and trade histories through [FINRA Fixed Income Data](#). Record identifiers, observation dates, price type and data permissions. Do not replace estimated risk exposures with unverified numbers. The embedded universe is the full fallback if an institutionally licensed inventory is unavailable.

Replace the embedded universe with a permitted bond inventory or evaluated-price snapshot. Add clean price and accrued interest, convert target market values to face amounts, impose issuer-level limits, and compare predicted tracking error with realized active returns over the next month.

11 Guided explorer

Keep Treasury yields unchanged and move corporate spreads; agencies move by one quarter as much. Compare the three candidate portfolios before recommending a trade. These are instantaneous mark-to-market scenarios, excluding carry and transaction costs.

The HTML control selects a precomputed scenario. In the PDF, use the preceding figures and tables and rerun the displayed notebook inputs to explore other values.